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Stadium Ticket Booking Cloud Infrastructure Project Report

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1. Introduction

This project presents the design and implementation of a secure, scalable, and well-governed **Stadium Ticket Booking Cloud Infrastructure** using Microsoft Azure. The objective was to simulate a real-world stadium management environment by implementing best practices in networking, identity management, storage security, policy enforcement, backup strategies, and role-based access control.

The infrastructure was designed to securely manage ticket booking operations, user access, storage services, virtual machines, and network communication while ensuring high availability and security.

For additional interactive visualization and deeper insights into the deployment, refer to:

- Architecture Diagram (Figma): [Azure Stadium Ticket Booking Cloud Infrastructure](#)
- Interactive Website: [Azure Stadium Ticket Booking Cloud Infrastructure Website](#)
- Screenshots PDF (Google Drive): [Azure Stadium Ticket Booking Cloud Infrastructure Screenshots](#)
- Project Recording(LOOM): [Azure Stadium Ticket Booking Cloud Infrastructure Screen Recording](#)

However, this document itself contains a complete explanation of the architecture and implementation.

2. Architecture Overview

The architecture is designed around security, scalability, and real-world stadium ticket management workflow. It consists of a multi-tier cloud infrastructure deployed within a single Azure Virtual Network, where different services are separated using dedicated subnets and secured using Network Security Groups (NSGs).

The system allows users to access the stadium ticket booking portal securely through Azure Virtual Machines and controlled network communication. Public-facing services are limited to only required components, while internal services remain protected inside private subnets. This approach reduces the attack surface and improves infrastructure security.

The project also implements identity and access management using Microsoft Entra ID, RBAC roles, Azure Policies, Storage Security, Backup Vault, and Load Balancer configurations to simulate an enterprise-level Azure deployment.

3. Infrastructure Components

The following Azure resources were deployed and configured for this project:

- 1 Resource Group (rg-stadium-sunil)
- 1 Virtual Network (CityNet)
- 3 Subnets (subnet-a, subnet-b, subnet-c)

- Network Security Groups (NSGs)
- Multiple Network Interfaces (NICs)
- 3 Linux Virtual Machines (vm-a-tickets, vm-b-app, vm-c-broadcast)
- 1 Storage Account (stsunilstadium123)
- 3 Blob Containers

- Azure Table Storage (StudentRecords)
- 1 Shared Access Signature (SAS) Token
- 1 Public IP

- Azure Backup Vault (stadiumBackupVault)
- 1 Recovery Services Vault
- 1 Backup Policy
- 1 Azure Policy Assignment
- Azure Entra ID Users and Groups
- RBAC Role Assignments

- Azure Load Balancer (stadiumLB)

All resources were deployed in the East Asia region to maintain consistency and simplify management.

4. Networking and Security

The Azure Virtual Network (CityNet) is divided into multiple isolated subnets to improve security and traffic management. Network Security Groups (NSGs) were configured at both subnet and virtual machine levels to control inbound and outbound traffic.

Custom security rules were implemented to allow only required communication between virtual machines. ICMP (Ping) validation was used to verify secure communication paths between services. Public access was restricted only to selected components using Public IP addresses and NSG filtering rules.

Ping validation results:

- vm-a-tickets → vm-b-app : Success
- vm-b-app → vm-c-broadcast : Success
- vm-a-tickets → vm-c-broadcast : Blocked by NSG

By limiting direct access between tiers, the architecture follows the principle of least privilege and reduces unnecessary exposure to attacks.

5. Identity and Access Management (IAM)

Microsoft Entra ID was used to create and manage users, groups, and role-based access control (RBAC) assignments for the project infrastructure.

Multiple users were created and categorized into administrative and operational groups to simulate enterprise-level access management. RBAC roles were assigned based on user responsibilities to ensure secure and controlled access to Azure resources.

Access Mapping:

- StadiumAdmins → Full administrative access
- Internal Users → Virtual machine and application access
- External Users → Limited access to selected resources

This RBAC-based implementation improves scalability, governance, and security management across the infrastructure.

6. Storage and Data Security

A dedicated Azure Storage Account (stsunilstadium123) was created to securely store ticket information, user records, broadcast files, and application-related data.

Blob Containers and Azure Table Storage (StudentRecords) were configured for structured and unstructured data storage. Shared Access Signature (SAS) tokens were generated to provide temporary and restricted access instead of exposing permanent storage account keys.

7. Azure Policy and Governance

Azure Policy was implemented to enforce governance and organizational standards across the infrastructure deployment.

Custom policies were assigned to restrict resource deployment locations and ensure that resources are created only in approved Azure regions such as Central India.

8. Backup and Recovery

An Azure Recovery Services Vault (stadiumBackupVault) was created to manage backup and recovery operations for the cloud infrastructure. Backup policies were configured to protect critical virtual machines and improve disaster recovery readiness.

The backup configuration helps ensure that virtual machine data and application resources can be restored in case of accidental deletion, system failure, or security incidents.

9. Infrastructure as Code (Terraform)

The complete infrastructure was deployed and managed using Microsoft Azure services through the Azure Portal and Azure CLI commands.

The project includes:

- Resource provisioning
- Virtual machine deployment
- Networking configuration

The deployment included networking, virtual machines, storage, backup, policy enforcement, and identity management.

10. Real-World Workflow

A user accesses the stadium ticket booking system through the web portal hosted on the Azure Virtual Machine (vm-a-tickets). The application processes ticket booking requests and securely communicates with the backend application server (vm-b-app) for ticket validation, user management, and event processing.

The backend services then store booking information, user records, and event-related data securely inside Azure Storage services and database-related resources. Network Security Groups (NSGs) control communication between virtual machines and restrict unauthorized access between network tiers.

Administrators manage infrastructure access through Microsoft Entra ID users, groups, and RBAC roles, while Azure Policy ensures governance and compliance across the deployment.

Azure Backup Vault protects important virtual machines and resources through backup configurations, helping improve disaster recovery and infrastructure reliability.

This workflow demonstrates how enterprise cloud services can securely manage networking, identity, storage, security, and operational management within Microsoft Azure.

11. Conclusion

This project demonstrates the successful deployment and management of a secure and scalable **Stadium Ticket Booking Cloud Infrastructure** using Microsoft Azure.

The implementation includes networking, virtual machines, storage management, backup configuration, Azure Policy enforcement, RBAC role assignments, Microsoft Entra ID users and groups, load balancing, and secure storage access using SAS tokens.

The project also demonstrates practical knowledge of Azure administration, networking, security, identity management, and cloud infrastructure deployment.

Overall, this project showcases how a real-world stadium ticket booking system can securely and efficiently manage cloud resources using Microsoft Azure while maintaining scalability, governance, and operational reliability.